

ECE 332 Fall 2025 — Exam 1 Answer Key (Alternative)*Variant: Q1(a) asks for total flux linkage $\Lambda = N\Phi$ instead of single-turn flux*[CS_{p.1}] = page 1 of cheat sheet [CS_{p.2}] = page 2**Two corrected errors from prior keys:****(1) Q1(c):** Exam writes “ $\omega = 2\text{ MHz}$ ” but MHz is a frequency unit. Must convert: $\omega = 2\pi f = 2\pi \times 2 \times 10^6 = 4\pi \times 10^6 \text{ rad/s}$.**(2) Q6:** Exam gives $f = \frac{2}{\pi} \text{ GHz}$ (not $\frac{2}{5} \text{ GHz}$). $\omega = 2\pi f = 2\pi \cdot \frac{2}{\pi} \times 10^9 = 4 \times 10^9 \text{ rad/s}$.

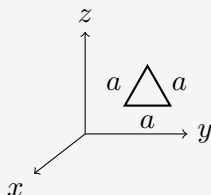
Question	Topic	Points	Page
Q1	Triangular loop EMF (flux linkage variant)	20	2
Q2	Moving wire (motional EMF)	12	3
Q3	Conduction & displacement current	12	4
Q4	Charge relaxation	10	5
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Q6	Lossy wave in distilled water	16	8
Total		100	

Question 1 — Triangular Loop

20 pts

Suppose an equilateral triangular conducting loop of N turns with sides of length a is located in the y - z plane with normal $d\mathbf{s} = \hat{x} ds$, as shown below, in a magnetic field

$$\mathbf{B}(t) = B_0(-2\hat{x} + 3\hat{y}) \sin(\omega t).$$



- (a) Find a formula for the total magnetic flux linkage Λ through N turns of the loop. (6 pts)
 (b) Find a formula for the emf induced in the loop at time t . (6 pts)
 (c) Determine the current flowing through a $1\text{ k}\Omega$ resistor incorporated in the loop when $N = 10$, $a = 8\text{ cm}$, $B_0 = 0.02\text{ T}$, $\omega = 2\text{ MHz}$, and $\omega t = \pi/4$, assuming that the resistance of the loop material is negligible. (8 pts)

Cheat Sheet Snippets (p.1)

EMF & FARADAY'S LAW

$$V_{\text{emf}} = -\frac{d\Lambda}{dt}, \quad \Lambda = N\Phi, \quad \Phi = \int_S \mathbf{B} \cdot d\mathbf{s}$$

DOT PRODUCTS

$$\hat{x} \cdot \hat{x} = 1, \quad \hat{y} \cdot \hat{x} = 0$$

 $\partial B / \partial t$ TABLE

$$\frac{d}{dt} \sin(\omega t) = \omega \cos(\omega t)$$

SURFACE ELEMENT

$$y\text{-}z \text{ plane} \Rightarrow d\mathbf{s} = \hat{x} da$$

EQUILATERAL TRIANGLE

$$\text{Area (side } a\text{): } A = \frac{\sqrt{3}}{4} a^2$$

CIRCUIT & WAVE + UNIT CIRCLE

$$\omega = 2\pi f; \quad \cos(\pi/4) = \frac{\sqrt{2}}{2}$$

- (a) First find the single-turn flux:

$$\Phi = \int_S B_0(-2\hat{x} + 3\hat{y}) \sin(\omega t) \cdot \hat{x} da = A(-2(\hat{x} \cdot \hat{x}) + 3(\hat{y} \cdot \hat{x})) B_0 \sin(\omega t) = -2 \cdot \frac{\sqrt{3}}{4} a^2 B_0 \sin(\omega t) = -\frac{\sqrt{3}}{2} B_0 a^2 \sin(\omega t)$$

The total flux linkage through N turns is $\Lambda = N\Phi$:

$$\Lambda = N\Phi = -\frac{\sqrt{3}}{2} N B_0 a^2 \sin(\omega t) \text{ V}\cdot\text{s}$$

- (b) Since Λ already contains N , differentiate directly — no extra N factor:

$$V_{\text{emf}} = -\frac{d\Lambda}{dt} = -\frac{d}{dt} \left(-\frac{\sqrt{3}}{2} N B_0 a^2 \sin(\omega t) \right) \quad V_{\text{emf}}(t) = \frac{\sqrt{3}}{2} N B_0 a^2 \omega \cos(\omega t) \text{ V}$$

- (c)

Exam says “ $\omega = 2\text{ MHz}$ ” $\Rightarrow$ convert: $\omega = 2\pi f = 2\pi(2 \times 10^6) = 4\pi \times 10^6 \text{ rad/s}$

$$|V_{\text{emf}}| = \frac{\sqrt{3}}{2} (10)(0.02)(0.08)^2 (4\pi \times 10^6) \underbrace{\cos(\pi/4)}_{\sqrt{2}/2} = 9.85 \times 10^3 \text{ V}$$

$$I = \frac{V_{\text{emf}}}{R} = \frac{9.85 \text{ kV}}{1000 \, \Omega} \quad \boxed{I \approx 9.85 \text{ A}}$$

Question 2 — Moving Wire

12 pts

A straight conducting wire of length $a = 15$ cm lying along the $\hat{y}$ direction moves with velocity $\mathbf{u} = 12\hat{x}$ cm/s in a magnetic field

$$\mathbf{B} = (0.1\hat{x} - 0.05\hat{z}) \text{ T.}$$

Find the emf between the two ends of the wire.

(12 pts)

Cheat Sheet Snippets (p.1)

EMF & FARADAY'S LAW (Motional) CROSS PRODUCT cycle

$$V_{\text{emf}} = \oint (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l}$$

$$\hat{x} \times \hat{z} = -\hat{y} \text{ (reversed cycle)}$$

$$d\mathbf{l} = \hat{y} dy \text{ along } \hat{y}\text{-wire}$$

$$\hat{x} \times \hat{x} = 0$$

$$a = 0.15 \text{ m, } \mathbf{u} = 0.12\hat{x} \text{ m/s.}$$

$$\mathbf{u} \times \mathbf{B} = 0.12\hat{x} \times (0.1\hat{x} - 0.05\hat{z}) = 0.012 \underbrace{(\hat{x} \times \hat{x})}_0 + (-0.006) \underbrace{(\hat{x} \times \hat{z})}_{-\hat{y}} = 6 \times 10^{-3} \hat{y} \text{ V/m}$$

$$V_{\text{emf}} = \int_0^{0.15} 6 \times 10^{-3} dy = 6 \times 10^{-3}(0.15)$$

$$V_{\text{emf}} = 900 \mu\text{V} = 0.9 \text{ mV}$$

Question 3 — Conduction & Displacement Current

12 pts

Suppose a total conduction current $I_c = 2 \sin(\omega t) \mu\text{A}$ flows through a column of distilled water with cross-sectional area A in a glass tube. Distilled water has conductivity $\sigma \approx 10^{-4} \text{ S/m}$ and relative permittivity $\epsilon_r \approx 26\pi$, and the current flowing through the glass is negligible.

(a) Use the relationship between the conduction current, the current density, the cross-sectional area, and Ohm's law to determine the electric field intensity E at a cross-section of the water. (6 pts)

(b) Determine the displacement current I_d through the cross-section when the angular frequency of I_c is $\omega = 10^9 \text{ rad s}^{-1}$ and $\omega t = \pi/3$. (6 pts)

Cheat Sheet Snippets (p.1)

DISPLACEMENT & CONDUCTION CURRENT UNIT CIRCLE

$$E = V(t)/d \quad I_c = \frac{\sigma A}{d} V(t) \quad J_c = \sigma E \quad \cos(\pi/3) = \frac{1}{2}$$

$$I_d = C \frac{dV}{dt} \quad C = \frac{\epsilon_r \epsilon_0 A}{d} \quad J_d = \epsilon \frac{\partial E}{\partial t}$$

(a)

$$E = \frac{V(t)}{d}, \quad I_c = \frac{\sigma A}{d} V(t) \Rightarrow V(t) = \frac{I_c d}{\sigma A} \quad E = \frac{I_c}{\sigma A} = \frac{2 \times 10^{-6} \sin(\omega t)}{10^{-4} \text{ A}} \quad \boxed{E = \frac{0.02}{A} \sin(\omega t) \text{ V/m}}$$

(b) From (a): $V(t) = E \cdot d = \frac{d \cdot 0.02}{A} \sin(\omega t)$.

$$I_d = \frac{\epsilon_r \epsilon_0 A}{d} \cdot \frac{dV}{dt} = \frac{\epsilon_r \epsilon_0 A}{d} \cdot \frac{d \cdot 0.02}{A} \cdot \omega \cos(\omega t) = \epsilon_r \epsilon_0 \cdot 0.02 \cdot \omega \cos(\omega t)$$

At $\omega t = \pi/3$, $\cos(\pi/3) = \frac{1}{2}$:

$$I_d = (26\pi)(8.85 \times 10^{-12})(0.02)(10^9)\left(\frac{1}{2}\right) \quad \boxed{I_d \approx 7.23 \text{ mA}}$$

Question 4 — Charge Relaxation

10 pts

Distilled water has a conductivity $\sigma \approx 10^{-4} \text{ S/m}$ and $\varepsilon_r \approx 26\pi$.

(a) Find the relaxation time constant for distilled water. (5 pts)

(b) How long does it take for an excess charge density in distilled water to decay to e^{-3} times its initial value? (5 pts)

Cheat Sheet Snippet (p.1 — CHARGE RELAXATION)

$$\tau_r = \frac{\varepsilon_r \varepsilon_0}{\sigma} \quad \rho_v(t) = \rho_{v0} e^{-t/\tau_r}$$

(a)

$$\tau_r = \frac{\varepsilon_r \varepsilon_0}{\sigma} = \frac{(26\pi)(8.85 \times 10^{-12})}{10^{-4}} \quad \boxed{\tau_r \approx 7.23 \mu\text{s}}$$

(b) Set $\rho_{v0} e^{-3} = \rho_{v0} e^{-t/\tau_r}$ and take $\ln$:

$$\ln e^{-3} = \frac{-t}{\tau_r} \Rightarrow t = -\tau_r \ln(e^{-3}) = 3\tau_r \quad \boxed{t = 3 \times 7.23 \mu\text{s} \approx 21.69 \mu\text{s}}$$

Question 5 — Lossless Plane Wave

30 pts

Suppose the electric field intensity of an EM wave in a nonconducting, source-free medium with $\epsilon_r = 6$ and $\mu_r = 1$ is

$$\mathbf{E}(z, t) = 4 \cos\left((6\pi \times 10^8 \text{ rad/s})t - kz - \frac{\pi}{6}\right) \hat{y} \text{ V/m.}$$

- (a) Determine ω , k , λ , and η . (10 pts)
 (b) Find $\tilde{\mathbf{E}}(z)$. (5 pts)
 (c) Find $\tilde{\mathbf{H}}(z)$. (5 pts)
 (d) Determine the magnetic field intensity $\mathbf{H}(z, t)$. (5 pts)
 (e) Determine the average power density that the EM wave carries. (5 pts)

Cheat Sheet Snippets

WAVE PARAMS (p.2 strip)

$$k = \omega \sqrt{\mu_r \mu_0 \epsilon_r \epsilon_0}, \quad \lambda = 2\pi/k$$

$$\eta = \eta_0 / \sqrt{\epsilon_r} = 377 / \sqrt{\epsilon_r}$$

PHASOR METHOD (p.1)

$$\mathbf{E} = \text{Re}\{\tilde{\mathbf{E}} e^{j\omega t}\}; \text{strip } e^{j\omega t}$$

PHASOR $\leftrightarrow$ TIME (p.2)

$$\text{Re}\{A e^{j(\omega t - \beta z + \phi)}\} = A \cos(\omega t - \beta z + \phi)$$

DIRECTION OF PROP. (p.2)

$$-kz \text{ in arg} \Rightarrow \hat{k} = +\hat{z}$$

$$\hat{k} \times \hat{E} \text{ TABLE (p.2): } +\hat{z}, \hat{y} \Rightarrow -\hat{x}$$

LOSSLESS E & H (p.2)

$$\tilde{\mathbf{H}} = \frac{1}{\eta}(\hat{k} \times \tilde{\mathbf{E}})$$

POYNTING lossless (p.2)

$$\mathbf{S}_{\text{av}} = \frac{|E_0|^2}{2\eta} \hat{k}$$

(a)

$$\omega = 6\pi \times 10^8 \text{ rad/s} \quad k = (6\pi \times 10^8) \sqrt{(4\pi \times 10^{-7})(6)(8.85 \times 10^{-12})} \quad k \approx 15.40 \text{ rad/m}$$

$$\lambda = \frac{2\pi}{15.40} \quad \lambda \approx 0.408 \text{ m} \quad \eta = \frac{377}{\sqrt{6}} \quad \eta \approx 153.91 \Omega$$

(b) Let $\phi = \pi/6$. Strip $e^{j\omega t}$: $\tilde{\mathbf{E}} = 4 e^{j(-kz - \phi)} \hat{y}$.

$$\tilde{\mathbf{E}}(z) = 4 e^{-j(15.40z + \pi/6)} \hat{y} \text{ V/m}$$

(c) $\hat{k} = +\hat{z}$; $\hat{z} \times \hat{y} = -\hat{x}$:

$$\tilde{\mathbf{H}} = \frac{1}{\eta}(\hat{z} \times \tilde{\mathbf{E}}) = -\frac{4}{153.91} e^{-j(15.40z + \pi/6)} \hat{x} \quad \tilde{\mathbf{H}}(z) \approx -25.99 \times 10^{-3} e^{-j(15.40z + \pi/6)} \hat{x} \text{ A/m}$$

Question 5 (continued)

(d)

$$\begin{aligned}\mathbf{H}(z, t) &= \operatorname{Re}\left\{-25.99 \times 10^{-3} e^{-j(15.40z + \pi/6)} \hat{x} \cdot e^{j\omega t}\right\} \\ &= \operatorname{Re}\left\{-25.99 \times 10^{-3} e^{j(\omega t - 15.40z - \pi/6)} \hat{x}\right\}\end{aligned}$$

$$\boxed{\mathbf{H}(z, t) = -25.99 \times 10^{-3} \cos(6\pi \times 10^8 t - 15.40 z - \frac{\pi}{6}) \hat{x} \text{ A/m}}$$

(e) Lossless $\Rightarrow$

$$\mathbf{S}_{\text{av}} = \frac{|E_0|^2}{2\eta} \hat{z} = \frac{4^2}{2(153.91)} \hat{z} \quad \boxed{\mathbf{S}_{\text{av}} \approx 51.98 \times 10^{-3} \hat{z} \text{ W/m}^2}$$

Question 6 — EM Wave in Distilled Water

16 pts

Consider a $\frac{2}{\pi}$ GHz EM wave traveling in distilled water, where $\sigma \approx 10^{-4}$ S/m, $\epsilon_r \approx 26\pi$, and $\mu_r = 1$. The wave travels downward into the water, where the $+z$ direction is down and $z = 0$ is just below the surface. The amplitude of the electric field at the surface is 4 V/m.

- (a) Find ϵ''/ϵ' . (6 pts)
 (b) What type of medium is distilled water at this frequency? Justify your answer. (5 pts)
 (c) Find the skin depth for distilled water at this frequency. (5 pts)

Cheat Sheet Snippets (p.2)

WAVE PARAMS strip

$$\omega = 2\pi f \quad \eta = 377/\sqrt{\epsilon_r}$$

SKIN DEPTH — LOW-LOSS MEDIUM

$$\delta_s = 2/(\sigma\eta) \text{ (from } \alpha = \sigma\eta/2)$$

LOSSY MEDIA — CLASSIFICATION

$$\epsilon''/\epsilon' = \sigma/(\omega\epsilon_r\epsilon_0)$$

$$< 0.01 \Rightarrow \text{low-loss dielectric}$$

$$\omega = 2\pi f = 2\pi \cdot \frac{2}{\pi} \times 10^9 = 4 \times 10^9 \text{ rad/s}$$

(a)

$$\frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega\epsilon_r\epsilon_0} = \frac{10^{-4}}{(4 \times 10^9)(26\pi)(8.85 \times 10^{-12})}$$

$$\frac{\epsilon''}{\epsilon'} \approx 34.58 \times 10^{-6}$$

(b) Since $34.58 \times 10^{-6} \ll 0.01$, at this frequency distilled water is a **low-loss dielectric**.

(c)

$$\eta = \frac{377}{\sqrt{26\pi}} = \frac{377}{9.038} \approx 41.71 \Omega \quad \delta_s = \frac{2}{\sigma\eta} = \frac{2}{(10^{-4})(41.71)} \quad \delta_s \approx 479.5 \text{ m}$$

Score (if perfect): Q1: 20 — Q2: 12 — Q3: 12 — Q4: 10 — Q5: 30 — Q6: 16 = **100/100**